



SIMATS ENGINEERING

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Course Code: CSA1709

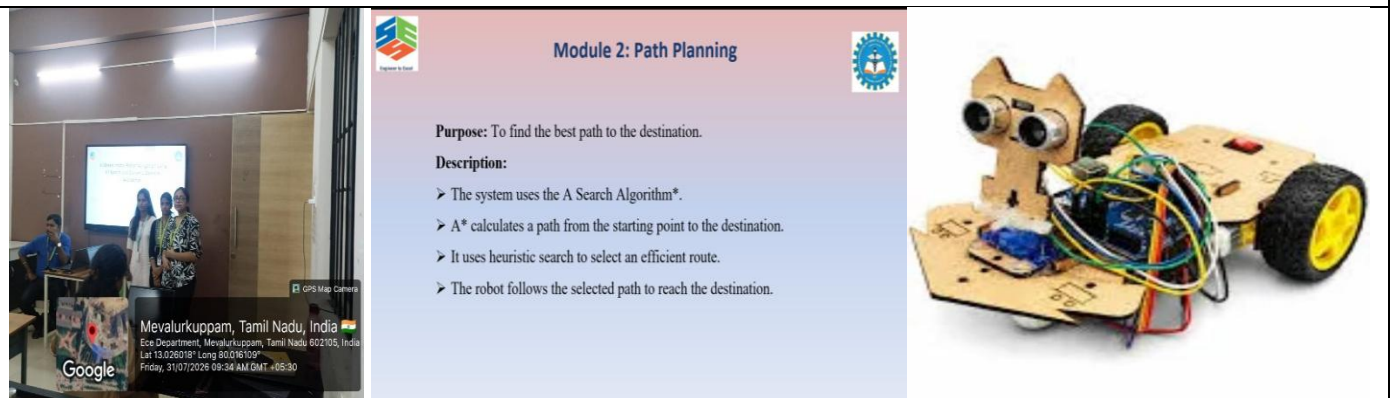
Slot: B

Course Name: Artificial Intelligence

Course Faculty: Dr.P.Rajaram

Project Title: AI Based Indoor Robot Navigation Using A* Search And Dynamic Obstacle Avoidance

Module Photographs:



Project Description: Module-2:Path Planning

The project “AI-Based Indoor Robot Navigation Using A Search and Dynamic Obstacle Avoidance”^{*} focuses on developing an intelligent robot navigation system for indoor environments such as classrooms, laboratories, offices, hospitals, and warehouses. The system enables the robot to find a safe and efficient route from its starting point to a specified destination while considering obstacles present in the environment.

In Module 2 – Path Planning, the A* (A-Star) search algorithm is used to determine the shortest and most efficient path. The indoor environment is represented as a grid or map containing free spaces and obstacles. A* evaluates the distance already travelled and the estimated distance to the destination using a heuristic function, allowing the robot to select an optimal path while avoiding static obstacles.

The project also incorporates dynamic obstacle avoidance to handle moving objects such as people, furniture, or other robots. When a new obstacle appears on the planned path, the system detects the change and dynamically replans an alternative route.

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